COURSE DESCRIPTION			
Program	**Automobile Engineering**		
Course Title	**Basic Automobile Engineering**		
Course Code	**1051**	**Semester**	**I**
Type of Course	**Practicum**	**Contact Hours**	**90**
Teaching Scheme (L: T:P:R)	**3:0:3:0**	**Credits**	**4.5**
CIE Marks	**40**	**ESE Marks**	**100**

Introduction

This course serves as the foundational pillar for aspiring Automobile Engineers, providing a holistic overview of the automotive ecosystem. It bridges the gap between basic sciences and core engineering by introducing students to the historical evolution of vehicles, the thermodynamic principles governing energy conversion, and the mechanical intricacies of Internal Combustion (IC) engines. Furthermore, the course addresses the "well-to-wheel" journey of automotive fuels, emphasizing safety standards and the environmental impact of emissions (BS-VI). Designed as a Practicum, it balances theoretical lectures with hands-on workshop exploration, ensuring students can identify, measure, and analyze vehicle components and thermodynamic systems from their first semester.

Course Objectives

1	To describe the evolution, industry players, and physical architecture of automobiles.
2	To explain fundamental thermodynamic laws and cycles related to energy conversion.
3	To identify internal combustion engine components and their operating principles.
4	To outline fuel properties, safety standards, and storage requirements.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Basic knowledge in science			Secondary Education

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Describe the historical development of automobiles, major Indian manufacturers, and the fundamental layouts of vehicle chassis and dimensions.	Understand
CO2	Explain the basic principles of thermodynamics, including system types, laws of energy conservation, and theoretical engine cycles.	Understand
CO3	Differentiate between internal engine components and the operating principles of 2-stroke, 4-stroke, Petrol, and Diesel engines.	Understand
CO4	Discuss the journey of fuel from crude oil to the vehicle tank, including fuel grades, safety protocols, and BS-VI standards.	Understand

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	2	-	-	-	-	-
CO2	3	2	-	-	-	-	-	-	-	-	-
CO3	3	2	-	2	-	-	-	-	2	-	-
CO4	3	-	-	-	3	3	3	2	-	-	-
Total	12	4	-	2	3	5	3	2	2	-	-
Correlation Level	3	2	-	2	3	2.5	3	2	2	-	-
Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"											

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	
3	0	3	0	6	90	4.5	20	60	80	20	40	60	140

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Syllabus – Major Topics

Module	Title	Major Topics	Instructional Hours	
			L	P
Module I	Introduction to Automobile	Evolution of Automobiles: History and development abroad and in India. Manufacturer Overview: Study of major concerns in India (Hero, Bajaj, Maruti Suzuki, Tata Motors, Ashok Leyland, Mahindra) including their plant locations and core products (2W, 3W, LMV, HMV). Chassis and Layout: Definition of chassis and body. Layout study of four-wheelers. Key Terminology & Dimensions: Geometric: Wheelbase, Wheel track, Front/Rear Overhang, Ground clearance. Weight: Kerb weight, Gross Vehicle Weight (GVW), and height of Center of Gravity (CG) . Engine Mounts: Purpose (vibration isolation), types (rigid, flexible, hydraulic), and selection criteria.	10	0
Module I	Introduction to Automobile	1. Structural Identification of Chassis and Body 2. Vehicle Layout Analysis - LMV 3. Vehicle Layout Analysis - HMV 4. Measurement of Vehicle Geometric Dimensions-LMV 5. Measurement of Vehicle Geometric Dimensions-HMV 6. Inspection and Selection of Engine Mounts	0	10
Module II	Basics of Thermodynamics	Foundations: Definition of system, surroundings, boundary, and universe. Types of systems (Open, Closed, Isolated). Properties & Equilibrium: Intensive vs. Extensive properties. Thermodynamic state and equilibrium. Laws of Thermodynamics: Zeroth Law- Concept of temperature and thermal equilibrium. First Law: Internal energy, enthalpy, and conservation of energy. Second & Third Laws: Entropy, heat engine efficiency, and absolute zero. Processes & Cycles: Isochoric, Isobaric, Isothermal, and Adiabatic processes (Basic only). P-V Diagrams of the Carnot, Otto, and Diesel cycles.	12	0
Module II	Basics of Thermodynamics	1. Identification of thermodynamic Systems in an Automobile. 2. Demonstration of the Zeroth Law and Thermal Equilibrium. 3. Study of Intensive and Extensive Properties of Engine Oil. 4. Experimental Verification of the First Law of Thermodynamics. 5. Experimental Verification of the Second Law of Thermodynamics. 6. Experimental Verification of the Third Law of Thermodynamics.	0	12
Module III	Engine Basics	Engine Components: Identification and function of the cylinder block, head, piston, connecting rod, crankshaft, and valves. Terminology: TDC, BDC, Stroke, Bore, Swept Volume, Clearance volume, Compression Ratio. Operating Principles: 4-Stroke Cycle: Suction, Compression, Power, and Exhaust (Petrol and Diesel). 2-Stroke Cycle: Comparison with 4-stroke, scavenging processes, and port timing. Comparison: Detailed technical comparison between SI (Spark Ignition) and CI (Compression Ignition) engines.	12	0
Module III	Engine Basics	1. Study of different tools used in Automobile Workshop 2. Identification of Major Engine Components 3. Measurement of Engine Geometrical Parameters 4. Demonstration of Four stroke cycle- Petrol engine 5. Demonstration of Four stroke cycle- Diesel engine 6. Demonstration of Two stroke cycle-Petrol engine	0	12

Module IV	Fuels	Fuels - Origin & Refining: Brief overview of Crude Oil extraction; Fractional Distillation process to derive Petrol (Gasoline) and Diesel. Storage & Handling: Safety protocols in fuel stations, underground storage tanks, and the vehicle fuel tank system (Fuel cap, filler neck, and venting). Fuel Specifications & Grades - Petrol Specifications: Octane Number , Volatility, and Calorific Value. Diesel Specifications: Cetane Number, Viscosity, and Sulfur content. Flash and Fire points. Grades & Standards: Understanding BS-VI (Bharat Stage) fuel norms; Bio-fuels (Ethanol blending in petrol). Layout of Fuel filling station.	11	0
Module IV	Fuels	1. Mapping the Fuel Journey (Flowchart) 2. Fuel Identification and Visual Inspection 3. Study of Vehicle Fuel Tank & Filling System 4. Study of safety Equipment at a Petrol Pump 5. Flash Point & Fire Point Demonstration (Safety Test) 6. Understanding BS-VI and Fuel Labeling	0	11

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (L/T)	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
Evolution of Automobile & Manufacturers	Identify the historical milestones and the core products/locations of major Indian automotive manufacturers.	L	CO1	Remember	2
Chassis & Layout	Classify the different types of vehicle chassis, bodies, and drivetrain layouts for 2, 3, and 4-wheelers.	L	CO1	Understand	3
Key Terminology	Define the fundamental vehicle dimensions and weight parameters like wheelbase, ground clearance, and GVW.	L	CO1	Remember	2
Engine Mounts	Explain the purpose and types of engine mounts used for vibration isolation in vehicles.	L	CO1	Understand	3
Structural Identification of Chassis and Body	Detail the major structural components of a vehicle	P	CO1	Understand	1.5
Vehicle Layout Analysis - LMV	Detail the arrangement of major aggregates in a Light Motor Vehicle (LMV) and explain the power flow	P	CO1	Understand	1.5
Vehicle Layout Analysis - HMV	Detail the arrangement of heavy-duty aggregates in an HMV and describe the configuration	P	CO1	Understand	1.5
Measurement of Vehicle Geometric Dimensions-LMV	Interpret the geometric dimensions of a Light Motor Vehicle (LMV) by identifying their measurement.	P	CO1	Understand	2
Measurement of Vehicle Geometric Dimensions-HMV	Interpret the geometric dimensions of a Light Motor Vehicle (HMV) by identifying their measurement	P	CO1	Understand	2
Inspection and Selection of Engine Mounts	Explain the function and selection criteria of different engine mounts by observing them.	P	CO1	Understand	1.5
Module II					
Thermodynamic Foundations	Define systems, surroundings, boundaries, and types of systems.	L	CO2	Remember	2
Properties & Equilibrium	Distinguish between intensive and extensive properties and describe state equilibrium.	L	CO2	Understand	3
Laws of Thermodynamics	Explain the Zeroth, First, Second, and Third laws of energy	L	CO2	Understand	4
Processes & Cycles	Discuss Isochoric, Isobaric, Isothermal, and Adiabatic processes.	L	CO2	Understand	3
Identification of Thermodynamic Systems in an Automobile	Classify various automotive components (like the radiator, fuel tank, and battery) as open, closed, or isolated systems based on mass and energy exchange.	P	CO2	Understand	2
Demonstration of the Zeroth Law and Thermal Equilibrium	Explain the concept of thermal equilibrium by observing temperature stabilization between a heat source (engine/coolant) and a measuring instrument.	P	CO2	Understand	2

Study of Intensive and Extensive Properties of Engine Oil	Distinguish between intensive properties (temperature, viscosity) and extensive properties (volume, mass) of automotive lubricants through observation.	P	CO2	Understand	2
Experimental Verification of the First Law of Thermodynamics	Interpret the law of conservation of energy by describing how chemical energy in fuel is converted into heat and mechanical work within a system.	P	CO2	Understand	2
Experimental Verification of the Second Law of Thermodynamics	Describe the necessity of a heat sink (radiator/exhaust) in an engine cycle and explain why 100% efficiency is physically impossible.	P	CO2	Understand	2
Experimental Verification of the Thrd Law of Thermodynamics	Discuss the theoretical concept of absolute zero and explain why it is impossible to reach minimum entropy in real-world automotive thermal processes	P	CO2	Understand	2
Module III					
Engine Components	Identify the major internal parts of an engine and their functions.	L	CO3	Remember	3
Engine Terminology	Define technical terms like TDC, BDC, Stroke, and Compression Ratio.	L	CO3	Remember	3
4-Stroke Operations (Petrol)	Describe the sequence of suction, compression, power, and exhaust.	L	CO3	Understand	2
4-Stroke Operations (Diesel)	Describe the sequence of suction, compression, power, and exhaust.	L	CO3	Understand	2
2-Stroke & Scavenging (Petrol)	Explain the two-stroke cycle and the scavenging process.	L	CO3	Understand	2
Study of different tools used in Automobile Workshop	Identify common hand tools, power tools, and precision measuring instruments, and describe their specific applications in automotive maintenance.	P	CO3	Understand	2
Identification of Major Engine Components	Classify internal and external engine parts (Piston, Crankshaft, Valves, etc.) and explain their individual functions within the engine assembly.	P	CO3	Understand	2
Measurement of Engine Geometrical Parameters	Interpret engine dimensions such as Bore, Stroke, and Clearance by identifying the correct measurement points using precision instruments.	P	CO3	Understand	2
Demonstration of Four stroke cycle- Petrol	Describe the sequence of the four strokes (Suction, Compression, Power, Exhaust) and explain the role of the spark plug in the SI engine cycle.	P	CO3	Understand	2
Demonstration of Four stroke cycle- Diesel	Explain the four-stroke operating principle in a CI engine and distinguish the fuel injection process from the spark ignition process.	P	CO3	Understand	2
Demonstration of Two stroke cycle- Petrol	Illustrate the two-stroke operating principle and describe the function of ports (Inlet, Transfer, Exhaust) and the scavenging process.	P	CO3	Understand	2
Module IV					
Origin & Refining of fuels	Outline the fractional distillation process of crude oil.	L	CO4	Remember	3
Storage & Handling of fuels	Describe safety protocols and vehicle fuel tank venting systems.	L	CO4	Understand	3
Specifications of fuels	Petrol Specifications: Octane Number , Volatility, and Calorific Value.	L	CO4	Remember	1
Specifications of fuels	Diesel Specifications: Cetane Number, Viscosity, and Sulfur content.	L	CO4	Remember	1
Specifications of fuels	Flash and Fire points.	L	CO4	Remember	1
Grades & Standards	Identify BS-VI norms	L	CO4	Remember	1
Lay out of fuel filling station	Picture graphically the layout of a fuel filling station	L	CO4	Understand	1
Mapping the Fuel Journey (Flowchart)	Summarize the stages of fuel processing from crude oil extraction to the retail outlet using a structured flowchart.	P	CO4	Understand	1
Fuel Identification and Visual Inspection	Distinguish between Petrol, Diesel, and Bio-fuels based on physical characteristics such as color, odor, and viscosity.	P	CO4	Understand	2

Study of Vehicle Fuel Tank & Filling System	Detail the components of a vehicle fuel system (filler neck, canister, venting) and describe how they prevent fuel leakage and vapor loss.	P	CO4	Understand	2
Study of safety Equipment at a Petrol Pump	Explain the purpose and operating principles of safety devices like fire extinguishers, sand buckets, and emergency cut-off switches at a filling station.	P	CO4	Understand	2
Flash Point and Fire point Demonstration (Safety Test)	Interpret the concept of "Flash Point" and "Fire Point" by observing the temperature at which fuel vapors ignite and discuss its importance for storage safety.	P	CO4	Understand	2
Understanding BS-VI and Fuel Labeling	Discuss the standard labeling and color-coding for BS-VI compliant fuels and explain the significance of emission norms in the Indian context.	P	CO4	Understand	2

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
	Module I	
1	Identify five different vehicles in your neighborhood and list their manufacturer, model name, and vehicle category (2W, 3W, LMV, or HTV).	6
2	Using a measuring tape, measure the Wheelbase and Ground Clearance of a family car or motorcycle and compare it with the official manufacturer specifications.	6
3	Open the hood of a car (with supervision) to locate the Engine Mounts and observe how they connect the engine to the chassis.	6
4	Draw a simple block diagram of a Front-Engine, Rear-Wheel Drive (FR) layout and label the major components (Engine, Gearbox, Propeller Shaft, Differential).	6
	Module II	
5	Analyze a household refrigerator and a pressure cooker; classify each as an Open, Closed, or Isolated system, giving reasons for your choice.	6
6	Using a thermometer, record the temperature of a cup of hot water every 5 minutes until it reaches Thermal Equilibrium with the room temperature	6
7	List 3 Intensive properties (e.g., Temperature) and 3 Extensive properties (e.g., Mass) you can observe while refilling a vehicle's coolant.	6
8	Watch an animation of the Otto Cycle and sketch the P-V diagram, labeling the points where "Suction" and "Compression" occur.	6
	Module III	
9	Visit a local garage to observe a dismantled engine and identify the Piston, Connecting Rod, and Crankshaft.	6
10	Use a mobile phone to record a video of a slow-motion animation of a 4-stroke engine and voice-over the four stages (Suction, Compression, Power, Exhaust).	6
11	Collect brochures of a Petrol car and its Diesel variant; compare their Compression Ratios and maximum power output.	6
12	Identify the Spark Plug in a petrol engine and the Fuel Injector in a diesel engine; note their locations	6
	Module IV	
13	Interview a driver or fuel station attendant to outline the steps involved in unloading fuel from a tanker truck to the underground storage tank	6
14	Visit a nearby Petrol Pump and list all the Safety Symbols and equipment (Fire extinguishers, sand buckets, "No Smoking" signs) you see.	6
15	Check the "Fuel Filler Flap" or the "Owner's Manual" of a modern vehicle to find the recommended Octane rating or BS-standard for that specific engine.	6
	Total Self-Learning hours	90

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Automobile Engineering (Vol. 1 & 2)	Dr. Kirpal Singh	Standard Publishers
2	Engineering Thermodynamics	P.K. Nag	McGraw Hill India
3	Elements of Fuel & Combustion Technology	O.P. Gupta	Khanna Publishing House

REFERENCE			
Sl. No.	Title	Author	Publication
1	A Textbook of Automobile Engineering	Er. S.K. Gupta	S. Chand Publishing
2	Basic Automobile Engineering	C.P. Nakra	Dhanpat Rai Publishing Co.
3	Applied Thermodynamics	T.D. Eastop & A. McConkey	Pearson Education
4	Internal Combustion Engines	V. Ganesan	McGraw Hill

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	Fundamentals of Automotive Systems - Course	I & III
2	Thermal Engineering: Basic and Applied - Course	II
3	NPTEL : Basic Thermodynamics (Mechanical Engineering)	II
4	NPTEL : NOC:Laws of Thermodynamics (Mechanical Engineering)	II
5	ARAI: The Automotive Research Association of India	I
6	SIAM	I
7	Home | Ministry of Petroleum and Natural Gas | Government of India	IV

Major Equipment/Instruments/Components required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	Multi-cylinder 4-Stroke Petrol Engine	4-cylinder MPFI engine, mounted on a trolley with cooling and fuel systems.	1
2	Multi-cylinder 4-Stroke Diesel Engine	4-cylinder CRDI engine, mounted on a trolley with battery and fuel tank.	1
3	2-Stroke Petrol Engine	Single-cylinder, air-cooled engine.	1
4	LMV Chassis	Fully functional LMV Chassis	1
5	HMV Chassis	Fully functional HMV Chassis	1

Major Software required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
	NIL		

Assessment Methodology – Practicum

Mode	Attendance	CA1	CE	CA2	CA3	CA4	CA5	ESE	
		Average of Self learning activities from each module	Continuous evaluation	Practical test (first half of experiments)	Practical test (second half of experiments)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)	Lab Examination (internal)
Duration				3 hours	3 hours	2 hours	2 hours	2.5 hours	3 hours
Exam mark		10	50	40	40	50	50	60	40
Converted to				10	10	10	10		
Marks	5	5	10	10		10		60	40
Total	40							100	
Tentative schedule	End of semester	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	End of semester	

Scheme of Evaluation - Practical (Continuous Internal Evaluation)

Sl. No.	Criteria	Description	Marks
1	Preparation & Punctuality	Attendance, timely submission of record, pre-lab preparation, understanding of experiment objectives and theory.	10

2	Experimental Setup & Procedure	Ability to identify components/equipment, correct circuit connections/setup, systematic and safe execution of the procedure.	10
3	Observation & Data Recording	Accuracy and neatness of readings, correct use of units, tabulation, and systematic recording of results.	5
4	Analysis & Result Interpretation	Proper calculations, graphical representation, result verification, discussion, and logical conclusion.	10
5	Viva Voce / Concept Understanding	Clarity of theoretical concepts, explanation of procedure, ability to answer questions related to the experiment.	10
6	Workmanship, Discipline & Teamwork	Laboratory discipline, care of instruments, teamwork, attitude, and safety practices.	5
Total			50

Scheme of Evaluation - Practical Test

Category	Things to evaluate	Marks
Write-up / Procedure (before experiment)	Aim, circuit diagram/flowchart, stepwise procedure.	10
Experiment Setup & Execution	Correct circuit connections / coding / setup. Handling of instruments/software. Accuracy of execution.	10
Observation & Result / Output	Proper tabulation of readings. Calculations/graphs/program output. Correctness of result / expected outcome.	8
Viva Voce	Conceptual understanding of experiment. Related theoretical knowledge.	8
Record	Completion & neatness of practical record. Proper certification by faculty.	4
Total		40

Series Examination Pattern- Theory

Time	Module (Series 1/ Series 2)	Part A	Marks	Part B	Marks	Part C (With choice)	Marks	Total
2 hours	Any two modules	2	1	3	3	2	7	25
	Any two modules	2	1	3	3	2	7	25
	Number	4		6		4		50

End Semester Examination Pattern of Question Paper - Theory

Duration	Module	TYPE OF QUESTIONS						
		Part A (2 Questions from each module)		Part B 2 out of 3 questions from each module)		Part C (1 set questions with choice from each module)		Total Marks
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (3 marks each)	No. of Questions	Marks (7 marks each)	
2.5 hours	1	2	2	3	6	1 set	7	15
	2	2	2	3	6	1 set	7	15
	3	2	2	3	6	1 set	7	15
	4	2	2	3	6	1 set	7	15
		8	8	12	24	4 sets	28	60